

GNOmE: Graph Networks for Mechanistic Explicability

Target Transformer (e.g. GPT-2, 12L × 12H)

Layer 1

Attention + MLP

Layer 2

Attention + MLP

Layer 3

Attention + MLP

Layer 4

Attention + MLP

Layer 5

Attention + MLP

Step 1: Graph

Single forward pass
Jacobians contributions

Step 2: Computation

156 nodes, edges via
GPT-2 (IOI)

Step 3: GNN

Node features →
Readout (e.g. softmax)

Step 4:

Importance

Query Complexity: $O(1)$

vs $O(N^2)$ for Path Patching
vs $O(N^2 \cdot T)$ for ACDC

Cross-Task Transfer: $r=0.954$

IOI → Induction transfer
Impossible for patching methods

GPT-2 Scale: 156 nodes

Full model extraction
IOI circuit recovery 3/7

vs Anthropic Circuit Tracing (2025): Backward Jacobian attribution → GNOmE uses forward-pass contribution vectors